SCIENCECLAW, MIT MEDIA LAB

Null Hypothesis

Can a collective of AI agents, each seeing one slice of the evidence and coordinating with no planner, find drug targets better than one agent that sees everything?

Team	Karuniya Premnath, Karishma Premnath, Dheepak Karan
Repository	github.com/karunprem45/natures-knockouts (public)
Submission tag	submission (commit 2307265)
Pre-registration	prereg (commit 12b8c58), set before any arm ran
Pre-registered headline	Collective vs single agent: delta AUROC +0.008 (95% CI -0.074 to +0.103), a null result
Strongest finding	Coordination penalty: A3 minus isolated specialists = -0.071 (95% CI -0.139 to -0.013)
Discovery	ACVR1C, a panel negative with published protective loss of function (54% lower T2D risk), surfaced blind
Tags	prereg, results-raw, results-final, submission, prereg-v2

This report summarises everything built and measured during the hackathon: the idea, the system, the data, every experiment and its result, how the work was kept honest, which platforms were used, and who did what.

1. Summary

Some people are born with a broken copy of a gene and are healthier for it. Loss of **PCSK9** means lifelong low LDL cholesterol, far fewer heart attacks, and eventually a drug (evolocumab). Finding the next such gene means combining evidence that lives in separate places: rare-variant burden studies, GWAS, population tolerance data and effects on other traits.

We built a decentralised collective of specialist agents for exactly that split. Each agent can open only its own folder of evidence; they coordinate through a shared board with no central planner. We tested it against a single agent that sees everything and a strong scripted baseline, on a blind panel of 40 genes frozen in git before any experiment ran.

Result	Value
Pre-registered headline: A3 (collective) minus A1 (single agent)	**+0.008 AUROC** (95% CI -0.074 to +0.103): a tie
A3 against chance (1,000 label shuffles)	permutation p = 0.003
Coordination penalty: A3 minus round 0 (secondary)	**-0.071** (-0.139 to -0.013): CI excludes zero
Isolated specialists minus single agent (secondary)	+0.080 (-0.004 to +0.173); +0.037 on GPT-5 mini
A3 + NVIDIA Evo 2 Variant agent (extension)	+0.038 AUROC over A3 (CI includes zero)
A2 centralized planner (extension)	Planner consulted everyone, so it matched round 0: a ~0.03 noise floor
Calibrated skeptic on a fresh panel (prereg-v2)	+0.004 (-0.032 to +0.042): no improvement
Top-ranked negative	ACVR1C: published 54% lower type 2 diabetes risk in loss-of-function carriers

Overall completion: the full BUILD.md plan (Phases 0-7) is done, plus the reusable skills, two platform extensions (NVIDIA Evo 2, OpenAI robustness) and every task in the final-gaps plan: the ACVR1C case file, the secondary comparisons, the deck rewrite, a centralized arm, a reproducibility check, and a pre-registered test of a calibrated skeptic on a fresh panel. Nebius was dropped because no API key was available to our code; Stage 2 binder design was out of scope by design.

2. The system

Agent	Sees only	Question it answers
Burden	In-scope rare-variant burden rows	Do carriers get less disease?
GWAS	In-scope fine-mapped credible sets	Does a second study type agree?
Tolerance	LOEUF, pLI, DepMap essentiality	Can a human live without it?
Safety	Out-of-scope traits; Mendelian disease only as categories	Does breaking it harm elsewhere?
Skeptic	Only the other agents' verdicts (no data)	Is this evidence trustworthy?
Variant (extension)	Lead variant, scored by NVIDIA Evo 2 40B	Is the variant really disruptive?

- **Round 0:** the four specialists post independent verdicts and broadcast needs for evidence they cannot see.
- **Round 1:** each agent inspects the board and bids on the open need it can answer, ranked by the organizers' formula: pressure = 2 x novelty + centrality + 0.5 x depth + 0.2 x age.
- **Round 2:** the skeptic posts typed challenges (underpowered, single cohort, single ancestry, biomarker only, direction conflict, essential gene); agents that received answers or challenges may revise.
- **Scoring** is pure Python, never the model: the latest verdict per channel, revisions supersede parents, a challenged channel counts half.

- Every artifact carries a UUID, a content hash and parent links, so each gene has a full provenance graph. Nothing in the loop assigns work.

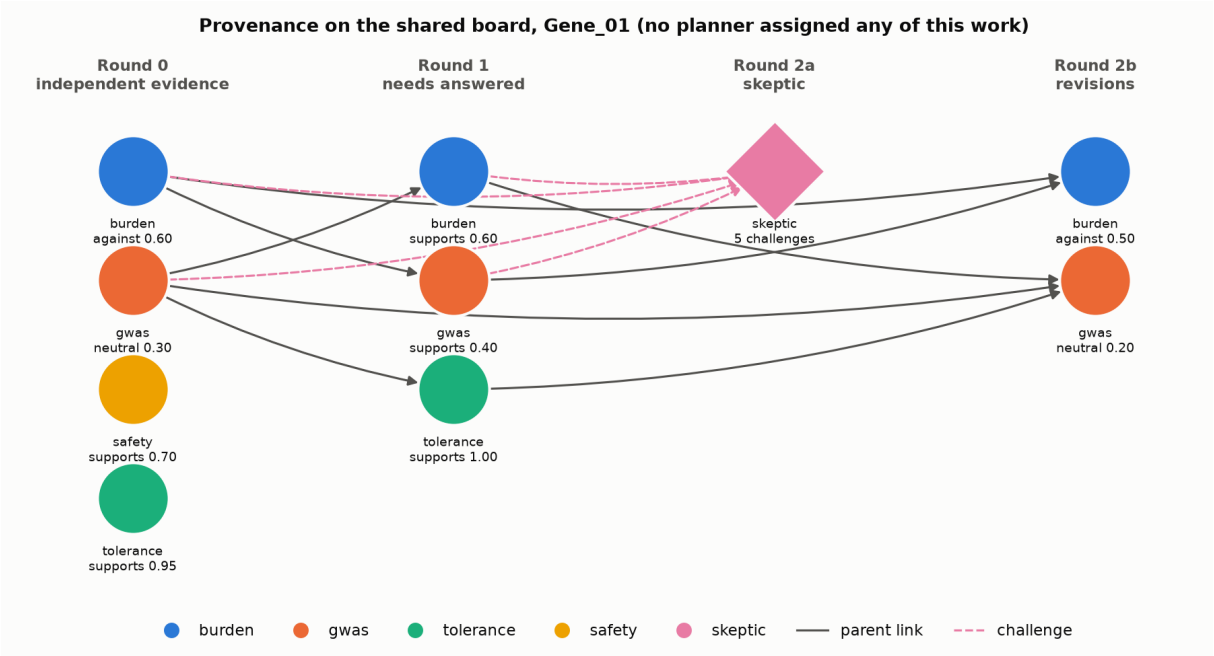


Figure 1. Provenance for Gene_01: four round-0 verdicts, three needs answered by other agents, one skeptic artifact with five challenges, and two revisions.

3. Data and the blind panel

- Source: Open Targets Platform release 26.09 (GraphQL), plus Ensembl REST for reference DNA.
- Candidate pool: 118 genes with in-scope, protective loss-of-function burden evidence (lipids, cardiovascular, metabolic, liver).
- Panel: 12 known targets (PCSK9, ANGPTL3, APOC3, ANGPTL4, NPC1L1, ASGR1, SLC30A8, MAP3K15, HSD17B13, CIDEB, GPR75, INHBE) and 28 hard negatives sampled with a fixed seed from 89 eligible genes.
- Genes are coded Gene_01 to Gene_40; labels live in a folder no agent code can read.
- **Leak found and closed:** in this Open Targets release, drug (ChEMBL) evidence is named *clinical_precedence*. It is routed only to the labels folder, and a test checks it never appears in a view.
- **Identity scrub:** 177 rows naming a gene or its own protein were dropped; Mendelian disease names were later coarsened to categories after an agent recognised a gene from eponymous syndromes.
- **Isolation test:** 203 checks prove no view holds another channel's data, a gene symbol, an Ensembl ID or drug data. A planted leak makes it fail.

4. What we built, phase by phase

Phase	What was built	Checkpoint result
0 - Setup	Repo, three-person commit credit, connection checks (Gemini, Open Targets, Ensembl, Evo 2)	Open Targets and Ensembl pass; Gemini via Vertex AI after sign-in
1 - Data	Open Targets client, pool, per-gene pulls, labels, blind panel, per-agent views, isolation test	40 genes x 5 views, 202 isolation checks pass; prereg tag set
2 - Baselines	Features, scripted arms S1-S4, metrics with bootstrap and paired delta	S4 AUROC 0.750 (0.542-0.910)
3 - Agents	Single LLM gateway with identity and label guards, call budget, logging; four agents and skeptic	25 artifacts on 5 genes, 0 errors; guard caught a real leak
4 - Board	Needs index with pressure scoring, round loop, revisions, Python scoring	10 genes: 20 cross-agent fulfilments; skeptic moved 9 scores
5 - Runs	A1 and A3, 2 seeds each, 12 zero-cost ablations, crash-safe checkpoints	0 error artifacts after a logged rerun of rate-limited genes
6 - Evaluation	Arms table, paired headline, permutation null, five figures, findings	Headline +0.008; results frozen with hashes
7 - Slides	Deck, two-minute demo replay (no API calls), demo script	Six slides incl. extensions; make demo works offline
Extensions	Multi-provider LLM layer; NVIDIA Evo 2 Variant agent; GPT-5 mini robustness run	Evo 2 +0.038; isolated specialists replicate at 0.857
Finish	Reusable skills, README, CONTRIBUTORS, three PRs, public repo, submission tag	247 tests pass; submission tag 2307265

Problems hit and how they were handled

- The hackathon GCP organisation disallows API keys, and gemini-2.5-flash was not served: switched to gemini-3-flash-preview on Vertex AI with Application Default Credentials (amendment 1).
- An agent named a gene from eponymous diseases: gene names in outputs are now redacted and flagged, and Mendelian diseases are shown as categories (amendment 2).
- Gemini sometimes wrapped JSON in a list, wasting 18% of calls on re-asks: the parser now unwraps it.
- Running four arms at once hit rate limits: 16 gene-runs were rerun under one rule for every arm, logged.
- A race condition in the Variant agent mislabelled one flag across parallel genes: fixed, with a regression test.

5. Results (pre-registered)

40 genes, 12 positives (chance AUPRC 0.30). LLM arms are averaged over 2 seeds. Confidence intervals come from 1,000 gene resamples; deltas are paired (both arms scored on the same resample).

Arm	What it is	AUROC (95% CI)	AUPRC	Calls / gene
S1	LOEUF alone	0.720 (0.559-0.868)	0.435	0
S2	Burden strength alone	0.662 (0.487-0.827)	0.396	0
S3	GWAS strength alone	0.440 (0.270-0.587)	0.276	0
S4	All features, z-scored, equal weights	0.750 (0.542-0.910)	0.697	0
A1	Single agent, all views	0.774 (0.616-0.903)	0.653	3
A3	Collective on the board	0.782 (0.641-0.895)	0.630	10
A3 round 0	Isolated specialists (re-scored)	0.853 (0.729-0.948)	0.736	0 extra
A3 no skeptic	Challenges removed (re-scored)	0.844 (0.713-0.939)	0.727	0 extra
A3 no burden	Burden agent removed (re-scored)	0.732 (0.589-0.862)	0.545	0 extra

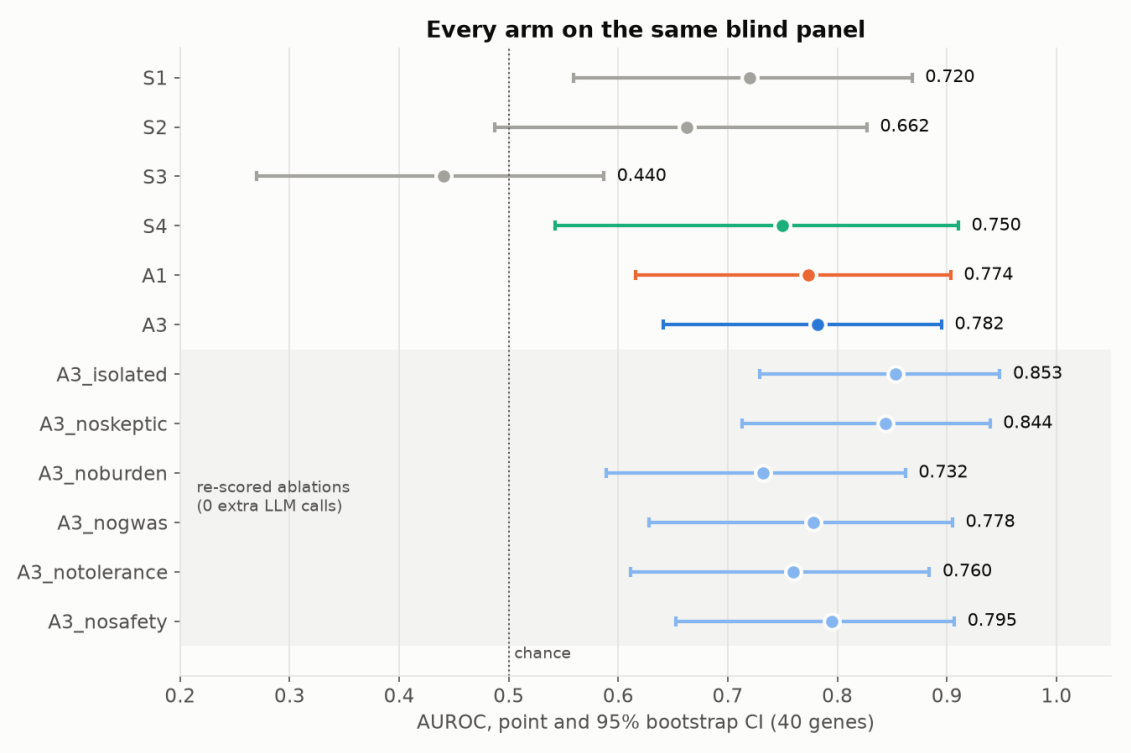


Figure 2. Every arm on the same blind panel, with 95% bootstrap intervals. Shaded rows are ablations re-scored from saved boards.

Why the coordination rounds hurt

The skeptic challenged the burden channel on 7 of 12 known targets and 21 of 28 negatives (seed 1), about the same rate whatever the truth, mostly as *biomarker only*. Halving the strongest channel almost everywhere pushed the known targets' mean rank from 10.7 to 12.5. Revisions barely moved verdicts. The headline stays A3 vs A1 as pre-registered; the ablations are reported as secondary findings and nothing was re-tuned after seeing them.

6. Extensions (after results-final, not pre-registered)

Run	Model(s)	Seeds	AUROC (95% CI)	Paired delta
A3 + Variant	Gemini 3 Flash + NVIDIA Evo 2 40B	2	0.820 (0.671-0.929)	+0.038 vs A3
A1	gpt-5-mini	1	0.820 (0.674-0.940)	-
A3	gpt-5-mini	1	0.751 (0.574-0.902)	-0.068 vs A1
A3 round 0	gpt-5-mini	1	0.857 (0.730-0.961)	+0.037 vs A1

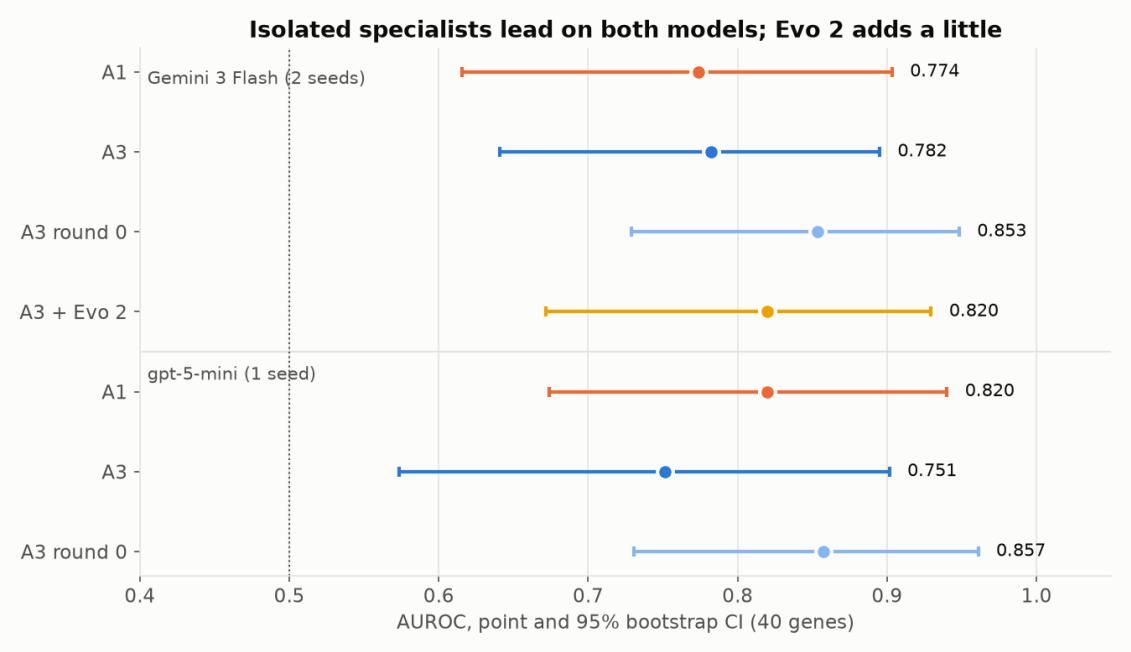


Figure 3. The same arms on two models, plus the Evo 2 Variant agent.

- **NVIDIA Evo 2:** one forward pass over 2,048 bp around the lead variant scores every possible base change in the window; the agent reports the variant's percentile and its log-likelihood change. Agents asked for it 77 times; Evo 2 answered 30 with no errors. Coordinates never reach a prompt.
- Most GWAS lead variants are non-coding tags (median percentile 59), so Evo 2 had little disruption signal to add. A fair test would feed it coding or burden-test variants on a fresh, pre-registered panel.
- **GPT-5 mini** rejects temperature 0, so it ran at its default temperature.
- **What replicates:** isolated specialists score about 0.85 on both models, and the full loop scores lower than its own round 0 on both.

7. Final gaps: post-freeze work (not pre-registered)

Everything in this section was added after results-final. The frozen chain was never touched: all 70 files under frozen/ and frozen_results/ were checked byte-identical at the start and end, and every SHA-256 verifies. Outputs live under exploratory/.

7.1 Secondary comparisons from frozen scores

Rubric comparison	Model	Contrast	Paired delta AUROC (95% CI)
Collaborative vs isolated	Gemini (2 seeds)	A3 - round 0	**-0.071 (-0.139 to -0.013)**, excludes 0
Collaborative vs isolated	GPT-5 mini (1 seed)	A3 - round 0	-0.106 (-0.239 to +0.008)
Multi-agent vs single-agent	Gemini	round 0 - A1	+0.080 (-0.004 to +0.173)
Multi-agent vs single-agent	GPT-5 mini	round 0 - A1	+0.037 (-0.084 to +0.158)
Pre-registered headline	Gemini	A3 - A1	+0.008 (-0.074 to +0.103)

The round-0 arm is a pre-registered ablation, but choosing these contrasts to highlight came after seeing results, with no multiplicity correction. Precision@12 and cost per arm were already in the frozen arms table (P@12 is pre-registered).

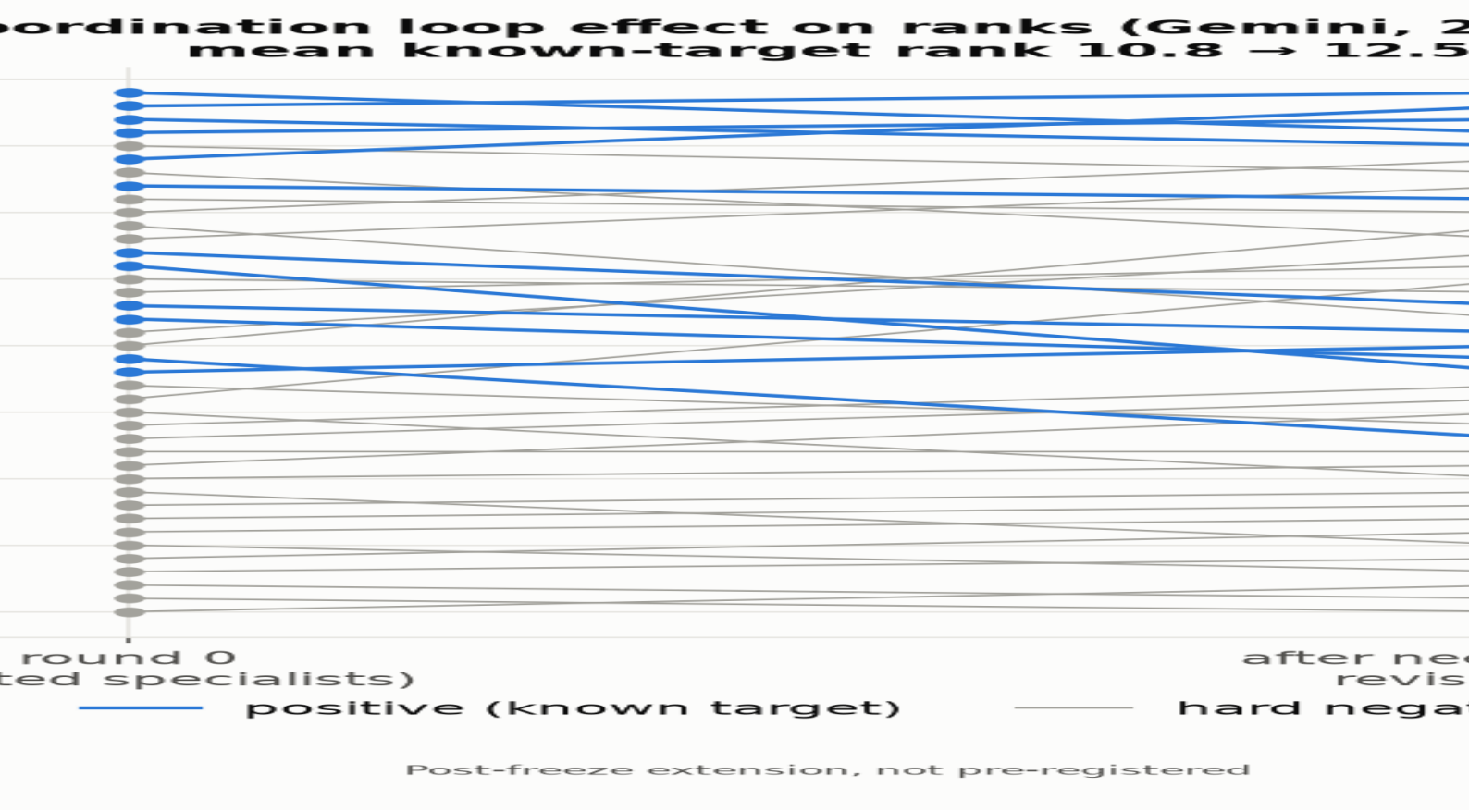


Figure 4. Per-gene rank from round 0 to the final A3 score (Gemini, 2 seeds). Known targets move from mean rank 10.8 to 12.5.

7.2 ACVR1C case file

A3 run	Rank of 40	Rank of 28 negatives
Seed-averaged	6	1
Seed 1	4	2
Seed 2	15	8

- Blind: 0 identity terms (symbol, Ensembl ID, 12 synonyms) and 0 drug terms in its 5 view files; its 5 isolation tests pass; no output flags.

- Seed 2's drop is the skeptic: it fired biomarker_only on burden and GWAS and halved both channels.
- Published evidence (Emdin et al., *Diabetes* 2019;68:226-234): carriers of predicted damaging ACVR1C variants had 54% lower type 2 diabetes risk (OR 0.46, 95% CI 0.27-0.81, exome sequencing of 55,516 people); in 405,569 UK Biobank participants, lower waist-to-hip ratio through ACVR1C meant 30% lower risk (OR 0.70, P = 5.6 x 10^{-13}).
- Framed as surfaced blind, not discovered: the 2019 evidence predates the run. ZNF229 and PKD1L3 stay hypotheses.

7.3 A2: centralized planner (1 seed)

Everything held constant except who decides the trajectory. A planner LLM saw only the gene code and one line per specialist, never evidence. It consulted all four specialists for all 40 genes, so A2 re-ran the round-0 configuration: 23 of 40 genes scored identically. A2 AUROC 0.871; A2 - round 0 = +0.028 (-0.008 to +0.090) is therefore a measured noise floor between two runs, not a centralization effect. Cost $0.60.

7.4 Reproducibility

- Clean clone: tests pass (8 data-dependent tests skip with a reason until the evidence is rebuilt); make demo runs with no keys and no network.
- verify_frozen: 63/63 hashes match, all 12 arms reproduce their AUROC from frozen scores, and round 0 re-scored from frozen boards matches exactly. Labels published in reproduce/ after results-final.
- REPRODUCE.md lists which steps run offline, which need network, and which need keys.

7.5 A calibrated skeptic on a fresh panel (prereg-v2)

Per-challenge-type discounts were fitted on panel 1 with one formula: multiplier = 1 - 0.5 x max(0, (rate on negatives - rate on positives) / rate on negatives). They were committed and tagged prereg-v2 before panel 2 existed.

Challenge type	Rate on positives	Rate on negatives	Multiplier
biomarker_only	0.83	0.84	1.00 (ignored)
direction_conflict	0.00	0.29	0.50
essential_gene	0.00	0.21	0.50
single_ancestry	0.25	0.48	0.76
single_cohort	0.08	0.36	0.62
underpowered	0.04	0.09	0.73

Panel 2 (12 shared positives, 28 fresh negatives)	AUROC (95% CI)
Calibrated skeptic	0.932 (0.835-0.997)
Flat 0.5 skeptic	0.927 (0.828-0.991)
Round 0 (isolated specialists)	0.954 (0.885-0.997)
Primary: calibrated - flat	**+0.004 (-0.032 to +0.042)**

Outcome, in the pre-registered wording: no improvement. The effect is far inside the 0.03 noise floor. Panel 2 is easier (round 0 at 0.954), so the flat skeptic cost only -0.027 and there was little to recover. The pre-registration set no threshold for improvement; the conservative rule (must beat the noise floor and exclude zero) was decided when results were read, and the report says so. 203 isolation checks passed on panel 2; one A3 run, 400 calls, 0 errors, $1.38.

8. How the work was kept honest

Safeguard	Detail
Pre-registration	prereg tag (12b8c58): panel, metrics and scoring weights frozen before any arm ran

Safeguard	Detail
Raw results frozen	results-raw tag (28a0e95): scores and boards hashed before any A1/A3 metric
Final results frozen	results-final tag (4729a6b); SHA-256 hashes for every output file
Blindness	Coded genes, per-agent folders, prompt identity guard, output redaction, 203-check isolation test
Fair comparison	Same model, same 10-call cap per gene, same scoring for A1 and A3; similar spend ($1.09 vs $1.32 per run)
Amendments	Two, both before any LLM arm ran: model change, and Mendelian diseases as categories
Reruns	Only API-error genes, one rule for every arm, logged in RERUN_LOG.md
Second pre-registration	prereg-v2 tagged before panel 2 existed; frozen chain verified untouched all night

Limitations, stated up front

- 40 genes: every interval is wide; S4, A1 and A3 overlap.
- The secondary contrasts were chosen after seeing results, with no multiplicity correction.
- The centralized arm's planner saw no evidence and consulted everyone, so decentralized vs centralized is not meaningfully tested.
- ACVR1C is the top negative only on the seed-averaged score (4th in seed 1, 15th in seed 2).
- Famous targets carry far more evidence rows, so any arm can partly score how well studied a gene is.
- Names, IDs and drug data are removed, but trait patterns remain (2 of 800 A3 artifacts still named a gene).
- Open Targets labels quantitative traits 'protective' from the sign of beta, so lower HDL reads as protective.
- Top-ranked negatives (ACVR1C, ZNF229, PKD1L3) are hypotheses, not validated targets.

9. Platforms used

Platform	Used for	Status
Google Cloud, Vertex AI (Gemini 3 Flash)	Every agent in every pre-registered arm	Used
NVIDIA build.nvidia.com (Evo 2 40B)	Variant agent	Used (free trial endpoint)
OpenAI (gpt-5-mini)	Robustness check on a second model	Used (hackathon credit)
Open Targets, Ensembl	Genetics evidence and reference DNA	Used
GitHub	Repository, tags, PRs, collaborators	Used (public)
NVIDIA Brev	Fallback GPU host for Evo 2	Not needed: hosted Evo 2 worked
Nebius	Planned open-model arm	Dropped: no API key usable by our code

Approximate spend: about $10.50 of Gemini (2,861 calls, including A2 and panel 2), about $1.20 of OpenAI credit, and nothing for Evo 2.

10. Team and contributions

Person	Role	Owned	Commits
Karuniya Premnath	Data	phase-0-probes, phase-1-data, fix-json-list, phase-7-slides, PR #1 (deck), deck rewrite and updates	11
Karishma Premnath	Agents	phase-3-agents, phase-3b-safety-coarsen, phase-4-board, explore-platforms (Evo 2), PR #2 (skills), A2 arm, reproducibility, panel-2 run	10
Dheepak Karan	Evaluation	phase-2-baselines, phase-5-arms, phase-6-eval, explore-openai, PR #3 (README), ACVR1C case and secondary comparisons, prereg-v2, this report	11

We worked as a mob on one machine; every commit lists the other two as co-authors. Authorship was reassigned by phase owner after the event; file contents are identical, the prereg hash is unchanged, and the original history is kept on archive/original-history.

11. Deliverables

Deliverable	Where
Code and tests (247 passing)	github.com/karunprem45/natures-knockouts
Pre-registered results	frozen_results/arms_table.md, FINDINGS.md (SHA-256 verified)
Post-freeze work	exploratory/: ACVR1C_CASE.md, explore_table.md, A2/, v2/prereg_v2.md, v2/RESULTS_V2.md
Six-slide deck	slides/deck.html (also published as a private artifact; share before judging)
Two-minute demo	make demo GENE=Gene_01 (replays a frozen board, no API calls); DEMO.md
Reproduction	REPRODUCE.md; exploratory.scripts.verify_frozen; reproduce/ labels
Reusable skills	skills/: blind-agent-views, needs-board-coordination, evo2-variant-score
Documentation	README.md, CONTRIBUTORS.md, BUILD.md, frozen/AMENDMENTS.md

Still to do before judging

- Share the deck from its Share menu.
- Rehearse the demo once, out loud.
- Discord, GitHub invites and the Infinite post (handled by a team member).
- Teammates: accept the GitHub invites and re-clone.

- After the event: revoke the NVIDIA and OpenAI keys that were pasted in chat.